Part A.

Carabello - Lopez

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1. $r = 0.55$

$LHR = 300$ W/cm $E = 200$ GPa $\alpha = 10 \times 10^{-6} \left[\frac{1}{K}\right]$

$k_f = 0.025$ W/cm-K $\nu = 0.35$

$$\sigma^* = \frac{\alpha E (T_0 - T_s)}{4(1-\nu)}$$

$$\sigma_\theta(\eta) = -\sigma^*(1 - 3\eta^2)$$

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I dont have any more time

$$\eta_1 = \sqrt{1 + \left(\frac{\sigma_f}{\frac{\sigma^*}{3}}\right)}$$

$$\Delta T = \frac{LHR}{4\pi k_f}$$

max stress

$$\eta = \frac{r}{R_f} = \frac{0.55}{}$$

max @ $r = R_f$

$$\Delta T = \frac{300}{4\pi(0.025)} = 954.93$$

$$\sigma^* = \frac{10 \times 10^{-6}(200)(\Delta T)}{4(1-0.35)}$$

$$\sigma^* = 0.734562$$

2.

thin

$$\sigma_\theta = \frac{PR}{\gamma} = \frac{(15)(0.55)}{0.05} = 165$$

$P = 15\,MPa$

$r = 0.55cm$

$\gamma = 0.05cm$

$$\sigma_z = \frac{PR}{2\gamma} = \frac{(15)(0.55)}{2(0.05)} = 82.5$$

6/8

$$\sigma_r = -\frac{P}{2} = -\frac{(15)}{2} = -7.5$$

$R_i = R_c - \left(\frac{t_c}{2}\right)$

$R_o = R_c + \left(\frac{t_c}{2}\right)$

$R_i = 0.55 - \left(\frac{0.05}{2}\right)$

$R_i = 0.525$

$R_o = 0.55 + \left(\frac{0.05}{2}\right)$

$R_o = 0.575$

$\frac{R_o}{R_i} = 1.09524$

thick

$$\sigma_r(r) = -P\,\frac{\left(\frac{R_o}{r}\right)^2 - 1}{\left(\frac{R_o}{R_i}\right)^2 - 1}$$

$$\sigma_\theta(r) = P\,\frac{\left(\frac{R_o}{r}\right)^2 + 1}{\left(\frac{R_o}{R_i}\right)^2 - 1}$$

$$\sigma_z(r) = P\,\frac{1}{\left(\frac{R_o}{R_i}\right)^2 - 1}$$

$$\sigma_r(r) = -15\,\frac{\left(\frac{0.575}{0.55}\right)^2 - 1}{(1.09524)^2 - 1} = -6.98884$$

- not sure what happened here,... math errors...

$$\sigma_\theta(r) = 15\,\frac{\left(\frac{0.575}{0.55}\right)^2 + 1}{(1.09524)^2 + 1} = 14.2732$$

$$\sigma_z(r) = 15\,\frac{1}{(1.09524)^2 - 1} = -12.5047$$

5/4

3.

$$T_{IC} - T_{OC} = \frac{LHR}{2\pi R_f h_{clad}} \qquad h_{clad} = \frac{K_{clad}}{t_{clad}} = \frac{0.15}{0.08} = 1.875\ cm$$

$$T_s - T_{IC} = \frac{LHR}{2\pi R_f h_{gap}} \qquad h_{gap} = \frac{0.003}{0.008} = 0.375$$

12/16

$$T_0 - T_s = \frac{LHR}{4\pi K_f} \qquad K_f = 0.03 \qquad T_{oc} = 550K$$

$$T_{IC} = \frac{200}{2\pi(0.48)(1.875)} + T_{oc} = 585.368\ K$$

LHR = 250 not 200

$$T_s = \frac{200}{2\pi(0.48)(0.375)} + T_{IC} = 762.207\ K$$

$$T_0 = \frac{200}{4\pi(0.03)} + T_s = 1,292.72\ K$$

gap calculation

$$① \quad \bar{R}_c = R_f + \delta_c + \left(\frac{t_c}{2}\right) = (0.48) + 0.06 + \left(\frac{0.06}{2}\right) - 0.57$$

t_g

$$② \quad \bar{T}_c = \frac{T_{CI} + T_{CO}}{2} = \frac{585.368 + 550K}{2} = 567.684\ K$$

$$③ \quad \Delta T_c = \bar{T}_c - T_0^{Ref} = 567.684 - 300 = 267.684$$

$$④ \quad \Delta R_c = \alpha_c \cdot \bar{R}_c + T_0^{Ref} = (10\times10^{-6})(0.57) + 300k = 300$$

?

$T_0 + T_s$

$$\bar{T}_f = \frac{(T_s + T_{co})}{2} = \frac{762.207 + 550\ K}{2} = 656.104$$

$$\Delta T_f = \bar{T}_f - T_0^{Ref} = 656.104 - 300 = 356.104$$

$$\Delta R_f = \alpha_f R_f \Delta T_f = (14\times10^{-6})(0.48)(356.104) = 0.002393$$

$$\Delta t_g = \Delta R_c - \Delta R_f = \cancel{300} - 0.002393 = 299.998$$

$$\Delta \delta_{gap} = \bar{R}_c \alpha_c (\bar{T}_c - T_0^{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref})$$

$$\Delta \delta_{gap} = (0.57)(10\times10^{-6})(567.684 - 300) - (0.48)(14\times10^{-6})(656.104 - 300)$$

$$\Delta \delta_{gap} = -0.000867$$

- right method, several mistakes

6/11

3. $\Delta \bar{R_c} = \alpha_c \bar{R_c} \times \Delta T_c$

$\Delta T_c = \bar{T_c} - T_0^{Ref} = 267.684$

$\Delta \bar{R_c} = (15\times10^{-6})(0.57) + 267.684 = 0.002289$

$\Delta R_f = \alpha_f \times R_f \times \Delta T_f$

$\Delta T_f = 356.104$

$\Delta R_f = (14\times10^{-6})(0.48)(356.104) = 0.002393$

$\Delta \partial_g = \Delta \bar{R_c} - \Delta R_f = 0.002289 - 0.002395 =$

$\Delta \partial_g = 0.000104$

7/11

4. cladding, hoop and radial

$\Delta T = 50K$

$\alpha_c = 15 \times 10^{-6}$ $\quad E = 100\ GPa$

$\nu = 0.34$ $\quad t_c = 0.06\ cm$ $\quad R_i = 0.55cm$

4/4

$$\sigma_r(r) = \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu}\left(\frac{r}{R_i} - 1\right)\left(1 - \frac{R_i}{2}\left(\frac{r}{R_i} - 1\right)\right)$$

not δ_c

$$\sigma_r(r) = \frac{1}{2}(50)\frac{(15\times10^{-6})(100GPa)}{1-0.34}\left(\frac{0.06}{0.55} - 1\right)\left(1 - \frac{0.55}{2}\left(\frac{0.06}{0.55} - 1\right)\right)$$

$\sigma_r(r) = -0.063022$ $\quad \sigma_r = 0$

$$\sigma_\theta(r) = \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu}\left(1 - 2\frac{R_i}{2}\left(\frac{r}{R_i} - 1\right)\right)$$

$$\sigma_\theta(r) = \frac{1}{2}(50)\frac{(15\times10^{-6})(100GPa)}{1-0.34}\left(1-2\left(\frac{0.55}{2}\left(\frac{0.06}{0.55} - 1\right)\right)\right)$$

r

$\sigma_\theta(r) = 0.0.84659$ - units?

8/11

5. $F = 6\times10^{13}$ fiss/cm³-s

$\Delta p_0 = 0.015$ $\rho(UO_2) = 10.97$ g/cc $1 FIMA = 950 \frac{MWD}{KgU}$

$\beta_D = 5 \frac{MWD}{KgU}$ $t = 10$ days $1 \text{ day} \approx 86400$ s

CD = 1 $N_A = 6.022\times10^{23}$

4/8

Number density uranium

$n_{UO_2} = 270$ g/mol

$$N_U = \left(\frac{\rho(UO_2) * N_A}{n_{UO_2}}\right) * \left(\frac{1U}{1UO_2}\right) = \frac{10.97 \text{ g/cm}^3 \times 6.022\times10^{23}}{270 \text{ g/mol}} = 2.446E^{22}$$

Burnup of the system

$$\beta = \frac{Ft}{N_U} = \frac{(6\times10^{13})(10 \text{ days} \times \frac{86400 \text{ s}}{1 \text{ day}})}{2.446E^{22}} = 0.0\cancel{0}0206$$

order of magnitude off...

Conversion to FIMA of β_D

$$\beta_D = 5 \frac{MWD}{KgU} \times \frac{1 \text{ FIMA}}{950 \frac{MWD}{KgU}} = 0.005263$$

strain calculation

$$\epsilon_D = \Delta p_0 \left(e^{\left(\frac{\beta \ln(0.01)}{(1)(\beta_D)}\right)} - 1 \right)$$

$$\epsilon_D = 0.015 \left(e^{\left(\frac{0.000206 \ln(0.01)}{0.005263}\right)} - 1 \right) = 0.004608$$

- wrong β

densification

9/4

Part B

6. Describing Plasticity and elasticity

4/4 Plasticity is the ability of a solid material to be able to undergo a permanent deformation. it is non reversible and this is seen in the shape of the response (typically by its applied forces). Meanwhile elasticity is the ability of a material to resist a distorting influence and return to the original shape when the influence is removed.

7. Define strain hardening

5/8 strain hardening is caused by dislocation pile ups this can occur after plastic deformation is unloaded changing the material.

-needed more here

8. fuel performace code requirements

3/3
- numerically model the temperature in the Fuel
- numerically model the stress in the cladding
- Consider the gap pressure, closure and heat transfer

9. grains in sintering

3/5 can form during the heat treatment process due to the heating and cooling of the material at extreme temperatures in its powder form, without melting. -needed more...

10 The Concept of mechanistic based on microstructure is based on knowing the materials properties during and after a mechanical process, since these properties obtained by the process can effect the fuel perfomce as imperfections to the material. It allows us to be able to predict a materials behavior.

4/8 -kind of... needed more

11. microstructure

5/8 microstructure is meant by the size of the lattices structures in the material, compactness, any bubbles, crevices or grains that could affect the materials preformace. -mostly

- casting is an example of a process that where a liquid material is poured into a mold and allowed to solidify. -Casting is fabrication, not processing
- these solidified microstructures usually are far from ideal due to its properties.

10/4

12. Diffusion / grains

4/4 It occurs faster along grain boundaries, due to that grain boundraies have more space than in the "perfect" lattice, making the atoms diffuse faster along these grain boundaries.

2/6 13. creep is the driving force of grain growth, defects in the material can reduce grain growth. → grain size also impacts by fission gas release swelling thermal conductivity

14. Densification of fuel is when the reduction in the surface area of pores, it occurs during the sintering process as a continuation. 3/6

- needed more

11/11